

Practice Final Fall 2023

1. Show these two logical expressions are equivalent:

$$B \vee C \vee (B \wedge C) \vee (A \wedge B \wedge C)$$

$$(\neg A \vee \neg B \vee \neg C) \wedge (B \vee C)$$

2. Find a linear recurrence for this sequence: 1,1,2,3,2,2,7,9,1,4,29,24

3. Modular arithmetic: Show there is no solution to $2x = 11 \pmod{18}$

4. What is the power set of $\{a, b, c\}$?

5. Make a Venn diagram showing $\neg(A \wedge B) \wedge \neg(A \wedge C) \wedge (\neg A \wedge \neg B \wedge \neg C)$

6. Evaluate $h(2,2)$ given $\forall n, m \geq 0$ (non-negative integers)

$$h(0, m) = 1$$

$$h(n+1, 0) = h(n, 2)$$

$$h(n+1, m+1) = 2 h(n+1, m)$$

7. System of Equations. Write in matrix form. Solve.

a)

$$5x + 7y + 9z = 2$$

$$2x + 5y + 5z = 8$$

$$5x + 5y + 6z = 4$$

$$x + y = 7$$

b)

$$6x + 5y + 6z = 2$$

$$8y + 8z = 1$$

$$7x + 2y + 8z = 5$$

c)

$$6x + 9y + 8z = 0$$

$$9x + 2y = 7$$

8. For $A = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix}$ what is the eigenvalue for the eigenvector $v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

9. Show that AA^t is a symmetric matrix.

10. In order to fit this bivariate data to the quantum model $y = a e^x + b e^{-x}$ using least squares on $Ax = b$, calculate A and b

x	y
-0.5	6.8
-0.25	4.6
0.	5.1
0.25	5.7
0.5	4.4

Out[]=

11. Given this matrix A

Out[]//MatrixForm=

$$\begin{pmatrix} 0.669066 & 0.450062 & 0.600905 & 0.275154 \\ 0.873963 & 0.328853 & 0.167913 & 0.273417 \\ 0.960839 & 0.527525 & 0.717818 & 0.0528061 \end{pmatrix}$$

a) Make an educated guess as to the matrix rank of A

b) Suppose this is the singular value decomposition of A.

$$\text{Out[]}= \begin{pmatrix} -0.54 & 0.29 & 0.79 \\ -0.49 & -0.87 & -0.012 \\ -0.68 & 0.39 & -0.61 \end{pmatrix} \cdot \begin{pmatrix} 1.9 & 0. & 0. & 0. \\ 0. & 0.39 & 0. & 0. \\ 0. & 0. & 0.2 & 0. \end{pmatrix} \cdot \begin{pmatrix} -0.76 & -0.4 & -0.47 & -0.17 \\ -0.48 & 0.13 & 0.79 & -0.35 \\ -0.37 & 0.13 & 0.15 & 0.91 \\ 0.22 & -0.9 & 0.35 & 0.16 \end{pmatrix}$$

Approximate A by $X \cdot Y$ where the rank of X and the rank of Y are both 1.

12. If A is 3 by 4 and B is 5 by 4, and C is the concatenation of A and B, then what are the dimensions of C?